SELECTION SORT

	i	j					
min = 0	10	9	8	7	6		min = 1
	0	1	2	3	4		
	i		j				
min = 1	10	9	8	7	6		min = 2
	0	1	2	3	4		
	i			j			
min = 2	10	9	8	7	6		min = 3
	0	1	2	3	4		
	i				j		
min = 3	10	9	8	7	6		min = 4
	0	1	2	3	4		
	i					j	
min = 4	6	9	8	7	10		swap(i, min)
	0	1	2	3	4		

	i	j					
min = 1	6	9	8	7	10		min = 2
	0	1	2	3	4		
	i		j				
min = 2	6	9	8	7	10		min = 3
	0	1	2	3	4		
	i			j			
min = 3	6	9	8	7	10		min = 3
	0	1	2	3	4		
	i				j		
min = 3	6	7	8	9	10		swap(i, min)
	0	1	2	3	4		

			i	j			
min = 2	6	7	8	9	10		min = 2
	0	1	2	3	4		
			i		j		
min = 2	6	7	8	9	10		min = 2
	0	1	2	3	4		
			i			j	
min = 2	6	7	8	9	10		swap(i, min)
	0	1	2	3	4		

				i	j		
min = 3	6	7	8	9	10		min = 3
	0	1	2	3	4		
				i		j	
min = 3	6	7	8	9	10		swap(i, min)
	0	1	2	3	4		

				i	j	
6	7	8	9	10		STOP
0	1	2	3	4		

INSERTION SORT

key = a[i]

	j i				
10	9	8	7	6	a[j]=a[j-1]
0	1	2	3	4	
	j i				
10	10	8	7	6	j--
0	1	2	3	4	
j	i				
9	10	8	7	6	a[j]=key
0	1	2	3	4	

key = a[i]

	j i				
9	10	8	7	6	a[j]=a[j-1]
0	1	2	3	4	
	j i				
9	10	10	7	6	j--
0	1	2	3	4	
j	i				
9	10	10	7	6	a[j]=a[j-1]
0	1	2	3	4	
j	i				
9	9	10	7	6	j--
0	1	2	3	4	
j	i				
8	9	10	7	6	a[j] = key
0	1	2	3	4	

key = a[i]

		j i			
8	9	10	7	6	a[j]=a[j-1]
0	1	2	3	4	
		j i			
8	9	10	10	6	j--
0	1	2	3	4	
		j	i		
8	9	10	10	6	a[j]=a[j-1]
0	1	2	3	4	
		j	i		
8	9	9	10	6	j--
0	1	2	3	4	
	j		i		
8	9	9	10	6	a[j]=a[j-1]
0	1	2	3	4	
	j		i		
8	8	9	10	6	j--
0	1	2	3	4	
j			i		
7	8	9	10	6	a[j]=key
0	1	2	3	4	

key = a[i]

			j i		
7	8	9	10	6	a[j]=a[j-1]
0	1	2	3	4	
			j i		
7	8	9	10	10	j--
0	1	2	3	4	
			j	i	
7	8	9	10	10	a[j]=a[j-1]
0	1	2	3	4	
			j	i	
7	8	9	9	10	j--
0	1	2	3	4	
		j		i	
7	8	9	9	10	a[j]=a[j-1]
0	1	2	3	4	
		j		i	
7	8	8	9	10	j--
0	1	2	3	4	
	j			i	
7	8	8	9	10	a[j]=a[j-1]
0	1	2	3	4	
	j			i	
7	7	8	9	10	j--
0	1	2	3	4	
j				i	
6	7	8	9	10	a[j]=key
0	1	2	3	4	

				i	
6	7	8	9	10	STOP
0	1	2	3	4	

BUBBLE SORT

i j					
10	9	8	7	6	swap(j, j+1)
0	1	2	3	4	
i j					
9	10	8	7	6	j++
0	1	2	3	4	
i	j				
9	10	8	7	6	swap(j, j+1)
0	1	2	3	4	
i	j				
9	8	10	7	6	j++
0	1	2	3	4	
i		j			
9	8	10	7	6	swap(j, j+1)
0	1	2	3	4	
i		j			
9	8	7	10	6	j++
0	1	2	3	4	
i			j		
9	8	7	10	6	swap(j, j+1)
0	1	2	3	4	
i			j		
9	8	7	6	10	j++
0	1	2	3	4	
i				j	
9	8	7	6	10	
0	1	2	3	4	

j	i				
9	8	7	6	10	swap(j, j+1)
0	1	2	3	4	
j	i				
8	9	7	6	10	j++
0	1	2	3	4	
i j					
8	9	7	6	10	swap(j, j+1)
0	1	2	3	4	
i j					
8	7	9	6	10	j++
0	1	2	3	4	
	i	j			
8	7	9	6	10	swap(j, j+1)
0	1	2	3	4	
	i	j			
8	7	6	9	10	j++
0	1	2	3	4	
	i		j		
8	7	6	9	10	
0	1	2	3	4	

j		i			
8	7	6	9	10	swap(j, j+1)
0	1	2	3	4	
j		i			
7	8	6	9	10	j++
0	1	2	3	4	
	j	i			
7	8	6	9	10	swap(j, j+1)
0	1	2	3	4	
	j	i			
7	6	8	9	10	j++
0	1	2	3	4	
		ij			
7	6	8	9	10	
0	1	2	3	4	

j			i		
7	6	8	9	10	swap(j, j+1)
0	1	2	3	4	
j			i		
6	7	8	9	10	j++
0	1	2	3	4	
	j		i		
6	7	8	9	10	
0	1	2	3	4	

				i	
6	7	8	9	10	STOP
0	1	2	3	4	

BUBBLE SORT ALTERNATIVE

j				i	
10	9	8	7	6	swap(j,j+1)
0	1	2	3	4	
j				i	
9	10	8	7	6	j++
0	1	2	3	4	
	j			i	
9	10	8	7	6	swap(j,j+1)
0	1	2	3	4	
	j			i	
9	8	10	7	6	j++
0	1	2	3	4	
		j		i	
9	8	10	7	6	swap(j,j+1)
0	1	2	3	4	
		j		i	
9	8	7	10	6	j++
0	1	2	3	4	
			j	i	
9	8	7	10	6	swap(j,j+1)
0	1	2	3	4	
			j	i	
9	8	7	6	10	j++
0	1	2	3	4	
				ij	
9	8	7	6	10	
0	1	2	3	4	

j			i		
9	8	7	6	10	swap(j,j+1)
0	1	2	3	4	
j			i		
8	9	7	6	10	j++
0	1	2	3	4	
	j		i		
8	9	7	6	10	swap(j,j+1)
0	1	2	3	4	
	j		i		
8	7	9	6	10	j++
0	1	2	3	4	
		j	i		
8	7	9	6	10	swap(j,j+1)
0	1	2	3	4	
		j	i		
8	7	6	9	10	j++
0	1	2	3	4	
			ij		
8	7	6	9	10	
0	1	2	3	4	

j		i			
8	7	6	9	10	swap(j,j+1)
0	1	2	3	4	
j		i			
7	8	6	9	10	j++
0	1	2	3	4	
j	i				
7	8	6	9	10	swap(j,j+1)
0	1	2	3	4	
j	i				
7	6	8	9	10	j++
0	1	2	3	4	
	ij				
7	6	8	9	10	
0	1	2	3	4	

j	i				
7	6	8	9	10	swap(j,j+1)
0	1	2	3	4	
j	i				
6	7	8	9	10	j++
0	1	2	3	4	
	ij				
6	7	8	9	10	
0	1	2	3	4	

i					
6	7	8	9	10	STOP
0	1	2	3	4	